THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The supplement of the complement of 20° is: 1

(a) 110° **(b)** 100° **(c)** 120° **(d)** 130°

2. Kaprekar's routine on 2111 reaches 6174 after how many steps? 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

3. In $\triangle ABC$, $\angle A = \angle B + \angle C$. Then $\angle A$ is: 1

(a) 90° **(b)** 120° **(c)** 60° **(d)** 45°

4. A pipe fills $\frac{2}{5}$ of a tank in 12 minutes. The time to fill the whole tank (minutes) is: 1

(a) 15 **(b)** 30 **(c)** 60 **(d)** $30\frac{1}{2}$

5. A tank holds 1,25,000 litres. How many such tanks hold 60,00,000 litres? 1

(a) 49 **(b)** 58 **(c)** 47 **(d)** 48

6. The value of 23 × 99 can be found as 23 × 100 - 23 = : 1

(a) 2254 **(b)** 2277 **(c)** 2300 **(d)** 2377

7. $(2.5 + 1.75) \times 0.4 =$ 1

(a) 1.8 **(b)** 1.7 **(c)** 2.7 **(d)** 1.6

8. If $n = 10$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 56 **(b)** 55 **(c)** 57 **(d)** 54

9. The reflex angle that goes with an angle of 70° (so that together they make a complete turn) is: 1

(a) 280° **(b)** 310° **(c)** 300° **(d)** 290°

10. The Collatz sequence from 12 needs how many steps to reach 1? 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 11

11. The angles of a triangle are $(2x)^\circ$, $(3x + 5)^\circ$ and $(4x - 50)^\circ$. The largest angle is: 1

(a) 100° **(b)** 70° **(c)** 90° **(d)** 80°

12. $\frac{1}{3}$ of a wall is already painted. Then $\frac{1}{2}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{6}$

13. The smallest number that must be added to 34,56,789 to get the next lakh is: 1

(a) 43,211 **(b)** 44,211 **(c)** 43,111 **(d)** 99,999

14. 12 × 38 + 12 × 62 is equal to 12 × ?. The missing number is: 1

(a) 112 **(b)** 99 **(c)** 101 **(d)** 100

15. Which of these is closest to 1? 1

(a) 0.89 **(b)** 1.08 **(c)** 0.95 **(d)** 1.2

16. Which expression is always even for a whole number n? 1

(a) $n + 2$ **(b)** $3n$ **(c)** $2n + 4$ **(d)** $2n + 1$

17. **Assertion (A):** The complement of 35° is 55°. 1

Reason (R): Two angles are complementary if their sum is 90°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** The sum of two odd numbers is even. 1

Reason (R): An odd number is of the form 2n + 1, so the sum of two odd numbers is 2(m + n + 1).

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** A triangle cannot be formed with sides 3 cm, 4 cm and 8 cm. 1

Reason (R): The sum of any two sides of a triangle must be greater than the third side.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$. 1

Reason (R): The product of two fractions is (product of numerators) / (product of denominators).

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. In $\triangle ABC$ and $\triangle PQR$ it is known that $AB = PQ = 5$ cm, $BC = QR = 6$ cm and $\angle A = \angle P = 50^\circ$. Are the triangles necessarily congruent? Give the rule or the reason. 2

22. Simplify $\left(\frac{5}{6} + \frac{1}{3}\right) \div \left(\frac{5}{6} - \frac{1}{3}\right) \times \frac{1}{2}$. 2

23. Which is greater: 98,76,543 or 98,67,543? By how much? 2

24. Use the distributive property to find 37 × 48 + 37 × 52. 2

25. Using each of the digits $2, 5, 8, 4$ exactly once, form the largest and the smallest numbers having two digits before and two digits after the decimal point. Find their difference. 2

26. Write the sum of three consecutive multiples of 2, starting with $2n$, and simplify it. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Arrange 12,34,567, 12,43,567, 12,34,765 in descending order and find the difference between the largest and the smallest. 3

28. Evaluate 12×15 and $(12 + 2) \times (15 - 2)$ and find the difference between the two values. 3

29. Two friends start from the same place and walk in opposite directions at 3.5 km/h and 4.5 km/h. How far apart are they after 2.5 hours? After how many hours will they be 24 km apart? 3

30. Write the sum of 3 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 7$. 3

31. Two supplementary angles are in the ratio $1 : 4$. Find the complement of the smaller angle and the supplement of the larger angle. 3

OR

Start with 1235 and apply Kaprekar's routine until 6174 appears. Show every step. 3

32. The perimeter of an isosceles triangle is 40 cm. Each equal side is 2 cm longer than the base. Find the sides and show that such a triangle exists. 3

OR

A cyclist rides at $\frac{9}{2}$ km per hour for 2 hours, then at $\frac{15}{4}$ km per hour for 4 hours. Find the total distance and the average speed for the whole journey. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Case study. Radha buys 2.5 m of cloth at ₹85.50 per metre. 5

(i) Find the cost of the cloth.
(ii) She pays ₹250. How much change does she get?
(iii) What would 4 m of this cloth cost?
(iv) Her mother has ₹513. How many metres of this cloth can she buy?

34. Case study. Sunita makes a row of triangles with matchsticks. One triangle needs 3 sticks, two triangles joined in a row need 5 sticks, and three triangles need 7 sticks. 5

(i) How many sticks are needed for 5 triangles?
(ii) Write an expression for the sticks for n triangles.
(iii) Find the sticks for 10 triangles.
(iv) She has 41 sticks. How many triangles can she make?

35. Priya has ₹x. Her brother has ₹15 more than twice her money. Write expressions for the brother's money and for their total money, and find both when $x = 40$. 5

36. A transversal cuts two parallel lines and two interior angles on the same side are in the ratio $2 : 3$. Find these two angles and say how many of the eight angles measure 72° and how many 108°. 5

OR

Find all two-digit numbers that are exactly 2 times the sum of their digits. 5

37. $\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given $\angle A = (3x + 10)^\circ$, $\angle D = (5x - 20)^\circ$, $AB = (2y + 4)$ cm and $DE = (3y - 6)$ cm. Find x, y, $\angle A$ and AB. If $\angle B = 40^\circ$, find $\angle F$. 5

OR

Pipe A fills a tank in 6 hours, pipe B in 12 hours, and a drain empties the full tank in 24 hours. Find the fraction filled in 1 hour with all three open, and the time to fill the empty tank. 5

38. A city has 4,20,000 houses. Each house uses 150 units of electricity per month. Find the units used by the city in one month and in a year (12 months). Write the yearly figure in words and round it to the nearest crore. 5

OR

A school fete: 50 children each buy 4 tickets at ₹12 each. 30 adults buy 3 tickets at ₹20 each. The organisers spend ₹700 on prizes and ₹500 on decoration. Write one expression for their profit and evaluate it. 5

END OF QUESTION PAPER **Total: 80 Marks**

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